Project Proposal

Thesis Title: Concise and Engaging Title

by

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1 Introduction

By training for next-word prediction in large datasets of texts, language models learn to encode linguistic properties that can be leveraged in a wide variety of downstream language-related tasks. In particular, transformer-based language models have become a cornerstone technology in Natural Language Processing (NLP) by advancing the state-of-the-art in many of its tasks. However, their data-hungry nature limits their applicability to the few languages that have sufficiently large datasets for language models to train on.

One solution to this limitation is to have language models train on texts written in a variety of languages. By doing so, such models pick up on near-universal linguistic properties, such as there being a distinction between nouns and verbs [7]; as well as properties shared only by subsets of languages, such as verbs preceding their subject. This information is then encoded in a shared semantic space that allows multilingual language-models to use information obtained from one language when processing another in a process called **cross-lingual transfer**. This is particularly useful in the low-resource setting: a language model may not have trained on enough data in Scottish Gaelic to have captured that verbs precede their subject in the language. However, if the model has been trained on enough texts in Modern Standard Arabic, it may well have managed to encode that verbs precede their subjects in that language, and consequently transfer that knowledge cross-lingually to better deal with texts in Scottish Gaelic.

The study of which languages exhibit which linguistic properties is called **linguistic typology** [4]. The ability of a model to transfer knowledge from a source language to a target language has been linked to the typological similarity between those languages [10, 14]. Choenni & Shutova [2] show that the multilingual language model M-BERT carves its semantic space into language-specific subspaces. That is, sentence embeddings in a given language tend to cluster around a language-specific centroid. Moreover, probing experiments reveal that typological features are encoded jointly across languages. For example, subtracting the Modern Standard Arabic centroid from a Scottish Gaelic sentence embedding would remove the information from the embedding that verbs precede their subject. This indicates overlapping subspaces for typologically similar languages, which could be what underlies the relation between typological similarity and successful cross-lingual transfer.

2 Related Work

The link between typological similarity and successful cross-lingual transfer is well established. Pires et al. [14] found that typological similarity correlates positively with zero-shot learning performance. While they attributed their results to lexical overlap between the source and target languages, Karthikeyan et al. [8] found the same results when the source and target languages had no lexical overlap at all. Instead, they demonstrated structural similarity between two languages to be of greater importance.

Typological relatedness has informed cross-lingual research in low-resource settings, too. Nooralahzadeh et al. [13] use a multi-stage approach to cross-lingual transfer with MAML where between the initial stage of fine-tuning on a large dataset to instill task-specific knowledge and few-shot learning to low-resource languages, they fine-tune on auxiliary languages to optimize the model parameters for quick adaptation. Choenni et al. [1] adapt this method and match auxiliary languages and low-resource languages based on typological similarity to further facilitate cross-lingual transfer.

Tenney et al. [15] set out to quantify where specific types of linguistic information are encoded in BERT, and found each consecutive layer to handle incrementally high-level linguistic information. That is, earlier layers handle concrete features like parts-of-speech tagging, whereas more abstract information like dependency relations and coreference resolution is dealt with in later layers. In the multilingual setting, Choenni & Shutova [2] found that M-BERT's internal representations have a clear typological organization, with the model encoding typological features explicitly and jointly across languages.

This joint encoding could be what underlies the relation between typological similarity and successful cross-lingual transfer. Fine-tuning an MLM on any given language could see it increase the importance of the encodings of the typological features of that language. As a result, after fine-tuning, the model's parameters would be close to what they would have been if the model had been fine-tuned to a typologically similar language.

3 Methodology

The methodology involves a three-stage fine-tuning setup outlined in Section 3.2, with the model being fine-tuned for dependency parsing in a variety of languages. The choice for dependency parsing was based on the availability of uniform datasets in a wide variety of languages through Universal Dependencies (UD) [12]. The selection of languages is clarified in Section 3.3, the specifics of the model are discussed in Section 3.1.

3.1 Model

The model used is derived from UDify, [9] which is itself based on M-BERT [5]. UDify incorporates a layer attention mechanism that scales each layer for its importance to the task at hand. That is, during fine-tuning, a weighted sum $\mathbf{r}_j$ is computed for input token j over all layers $i \in [1, ..., 12]$ as follows:

$$\mathbf{r}_j = \eta \sum_i \mathbf{U}_{i,j} \cdot \text{softmax}(\lambda)_i$$

where $\mathbf{U}_{i,j}$ is the output of layer i at token position j and $\mathbf{w}$ is a vector trained alongside the model such that $\mathbf{w}_i$ reflects the importance of layer i. Dependency arcs between tokens are scored using a biaffine attention classifier, [6] after which the optimal parse tree is decoded using the Chu-Liu/Egmonds algorithm. [3]

3.2 Fine-tuning setup

As stated, the fine-tuning setup follows three stages adapted from Choenni et al. [1]

In the first stage, the model M is fine-tuned on a German-language (*deu*) treebank to instill general knowledge of the task at hand. In the second stage, four copies of M are fine-tuned separately on treebanks of auxiliary languages $\ell \in \{nld swe ces, hun\}$[1] to obtain four instances of M^{ℓ}. Finally, each M^{ℓ} is fine-tuned on each low-resource language $\lambda \in \{gsw, fao, hsb, vep\}$,[2] resulting in 16 instances of $M^{\ell,\lambda}$. If time permits, both the second and third stage are ideally repeated across at least five random seeds.

3.3 Languages

Broadly speaking, language selection is based on their typological similarity and their prevalence in M-BERT's pre-training data as reported by Wu & Dredze. [17] Typological similarity is defined using the syntactic feature-vectors (`syntax_knn`) from the URIEL knowledge base. [11] In particular, each language is represented as a feature-vector, and similarity between languages is taken to be the cosine similarity between their respective vectors.

German was selected as one of the most prominent languages in M-BERT's pre-training data. Moreover, Turc et al. [16] found German (alongside Russian) to be the most effective language for cross-lingual transfer. The auxiliary languages all have a roughly similar amount of data in M-BERT's pre-training data. Their selection is motivated phylogenetically, as they each represent a different level of relatedness to German (Hungarian is unrelated, Czech is in the same family, Swedish is in the same branch, Dutch is in the same subbranch). However, a hierarchical clustering analysis of the `syntax_knn` vectors (Appendix A) found their selection to by typologically valid, too.

4 Experimental Setup

The goal of the project is twofold: Firstly, it aims to show how typology relates to model parameter updates in the fine-tuning stage; secondly, it aims to investigate the effect of typological distance on the success of cross-lingual transfer. Section 4.1 shows how the setup outlined in Section 3.2 serves to achieve these goals. Section 4.2 lists the parameters used, Table B in Appendix B lists the specific datasets used in the experiments.

4.1 Analyses

Q1 In the second stage, four model instances optimized for dependency parsing in German is fine-tuned to do dependency parsing in languages with varying levels of similarity to German. Intuitively, the German-language data in the preceding stage would have

[1] Dutch, Swedish, Czech, and Hungarian.
[2] Swiss German, Faroese, Upper Sorbian, and Veps.

prepared it for dependency parsing in the closely related Dutch more than in the unrelated Hungarian. H1 would suggest that the model instance that fine-tunes to Dutch sees less extensive parameter udpates than the one that fine-tunes to Hungarian. In addition, H2 suggests that the updates should primarily be to the earlier layers of the model for Dutch, whereas for Hungarian, they would be across all layers.

The fine-tuning setup includes learned scalar weights $\mathbf{w}_i$ that indicate the importance of each layer i. Following Tenney et al. [15], these can be used to detect where model updates took place for each ℓ. With $\Delta\mathbf{w}^{\ell} = \mathbf{w}^{\ell} - \mathbf{w}^{deu}$ being the shift in layer importance after fine-tuning on ℓ, the expectation is that $\Delta\mathbf{w}_i^{nld}$ is greater for layers i that deal with lexical information (i.e., the earlier ones [15]), but lower for layers i that deal with higher-level typological features, since those are generally shared between Dutch and German. On the other hand, the values of $\Delta\mathbf{w}^{hun}$ are expected to be more uniform, since Hungarian differs significantly from German both lexically and syntactically.

Q2 In the third stage, the model instances for languages ℓ are fine-tuned further to each the low-resource languages λ. Each $M^{\ell,\lambda}$ is then evaluated on the test set for λ, with performance measured as the Labeled Attachment Score (LAS). H3 suggests that greater cosine similarity $s(\ell, \lambda)$ should correspond to higher performance on λ's test set.

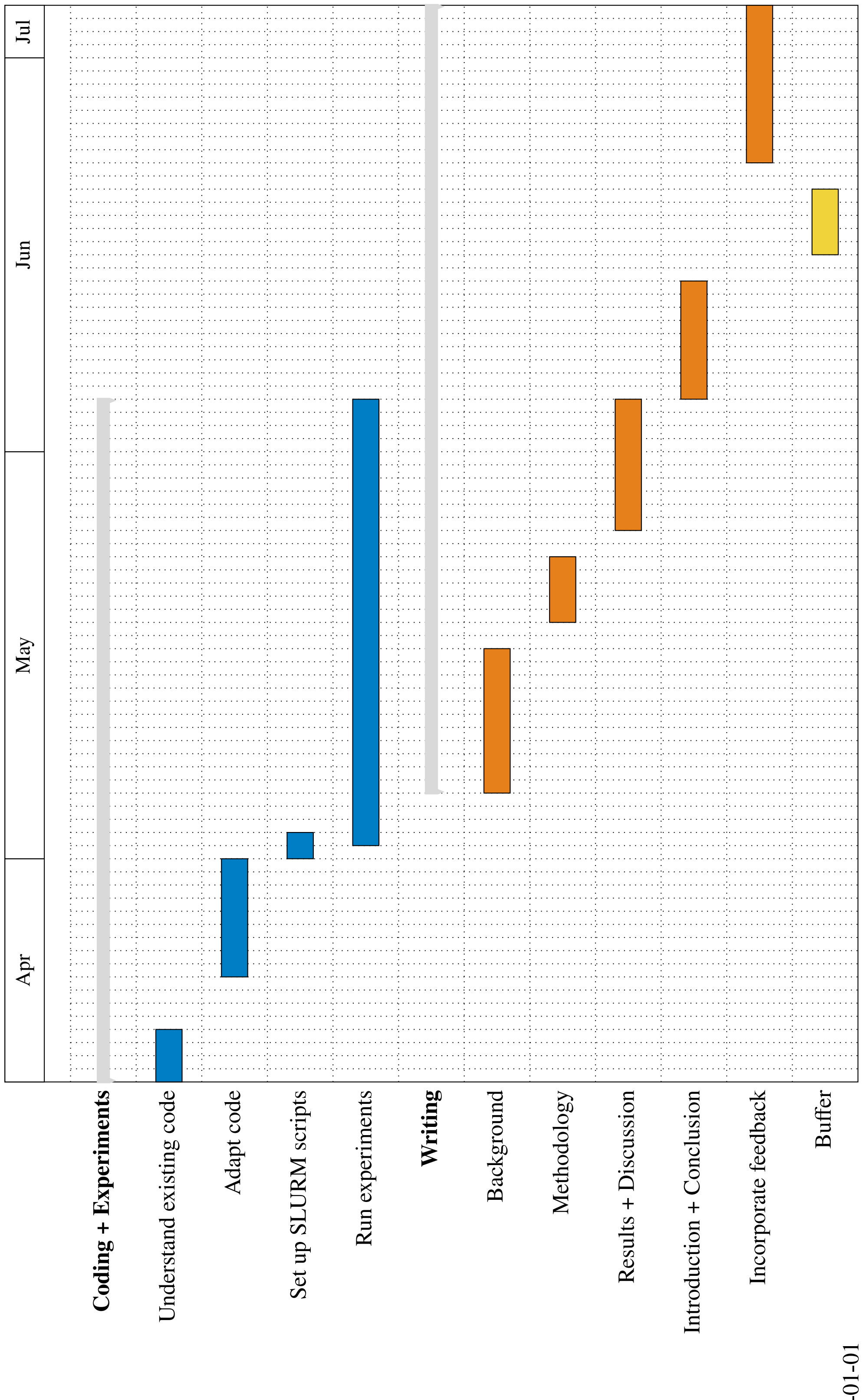
4.2 Parameters

The model is first fine-tuned to Part A[3] of the German training set. Choenni et al. [1] use the English EWT treebank with 12.5k samples in this stage and train for 60 epochs. To match the number of examples in this setup, training on the German treebank spans 11 epochs. In the second stage, the model instances are fine-tuned over 1 000 iterations with a batch size of 20 for all languages ℓ. In the third stage, the models are fine-tuned on 20 examples sampled randomly from each treebank of language λ. For the Upper Sorbian treebank, these are sampled from the train set. The other three treebanks do not split their data into separate sets; therefore, the sampled examples are removed from the dataset so as to prevent evaluating on them.

Following Choenni et al. [1], a cosine-based learning rate scheduler with 10 % warm-up and the Adam optimizer are used. For the language model, a learning rate of $1e-04$ is used, and for the classifier, a learning rate of $1e-03$ is used.

[3] The HDT treebank training data is split into various parts. With Part A being larger than the entire English-language treebank used by Choenni et al. [1], the decision was made to use just that part for training.

5 Planning


Apr
May
Jun
Jul
Coding + Experiments
Understand existing code
Adapt code
Set up SLURM scripts
Run experiments
Writing
Background
Methodology
Results + Discussion
Introduction + Conclusion
Incorporate feedback
Buffer
-01-01

References

1. Rochelle Choenni, Dan Garrette, and Ekaterina Shutova. Cross-lingual transfer with language-specific subnetworks for low-resource dependency parsing. *Computational Linguistics*, pages 613–641, September 2023.
2. Rochelle Choenni and Ekaterina Shutova. Investigating language relationships in multilingual sentence encoders through the lens of linguistic typology. *Computational Linguistics*, 48(3):635–672, September 2022.
3. Yoeng-Jin Chu. On the shortest arborescence of a directed graph. *Scientia Sinica*, 14:1396–1400, 1965.
4. William Croft. *Typology and Universals*. Cambridge University Press, 2002.
5. Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, June 2019. Association for Computational Linguistics.
6. Timothy Dozat and Christopher D Manning. Deep biaffine attention for neural dependency parsing. *arXiv preprint arXiv:1611.01734*, 2016.
7. Nicholas Evans and Stephen C Levinson. The myth of language universals: Language diversity and its importance for cognitive science. *Behavioral and brain sciences*, 32(5):429–448, 2009.
8. Karthikeyan K, Zihan Wang, Stephen Mayhew, and Dan Roth. Cross-lingual ability of multilingual BERT: an empirical study. *CoRR*, abs/1912.07840, 2019.
9. Dan Kondratyuk and Milan Straka. 75 languages, 1 model: Parsing Universal Dependencies universally. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2779–2795, Hong Kong, China, November 2019. Association for Computational Linguistics.
10. Anne Lauscher, Vinit Ravishankar, Ivan Vulić, and Goran Glavaš. From zero to hero: On the limitations of zero-shot language transfer with multilingual Transformers. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4483–4499, Online, November 2020. Association for Computational Linguistics.
11. Patrick Littell, David R. Mortensen, Ke Lin, Katherine Kairis, Carlisle Turner, and Lori Levin. URIEL and lang2vec: Representing languages as typological, geographical, and phylogenetic vectors. In Mirella Lapata, Phil Blunsom, and Alexander Koller, editors, *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 8–14, Valencia, Spain, April 2017. Association for Computational Linguistics.
12. Joakim Nivre, Marie-Catherine de Marneffe, Filip Ginter, Yoav Goldberg, Jan Hajič, Christopher D. Manning, Ryan McDonald, Slav Petrov, Sampo Pyysalo, Natalia Silveira, Reut Tsarfaty, and Daniel Zeman. Universal Dependencies v1: A multilingual treebank collection. In Nicoletta Calzolari, Khalid Choukri, Thierry Declerck, Sara Goggi, Marko Grobelnik, Bente Maegaard, Joseph Mariani, Helene Mazo, Asuncion Moreno, Jan Odijk, and Stelios Piperidis, editors, *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, pages 1659–1666, Portorož, Slovenia, May 2016. European Language Resources Association (ELRA).
13. Farhad Nooralahzadeh, Giannis Bekoulis, Johannes Bjerva, and Isabelle Augenstein. Zero-shot cross-lingual transfer with meta learning. In Bonnie Webber, Trevor Cohn, Yulan He,

and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4547–4562, Online, November 2020. Association for Computational Linguistics.

14. Telmo Pires, Eva Schlinger, and Dan Garrette. How multilingual is multilingual BERT? In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4996–5001, Florence, Italy, July 2019. Association for Computational Linguistics.
15. Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, Florence, Italy, July 2019. Association for Computational Linguistics.
16. Iulia Turc, Kenton Lee, Jacob Eisenstein, Ming-Wei Chang, and Kristina Toutanova. Revisiting the primacy of english in zero-shot cross-lingual transfer. *arXiv e-prints*, pages arXiv–2106, 2021.
17. Shijie Wu and Mark Dredze. Are all languages created equal in multilingual BERT? In Spandana Gella, Johannes Welbl, Marek Rei, Fabio Petroni, Patrick Lewis, Emma Strubell, Minjoon Seo, and Hannaneh Hajishirzi, editors, *Proceedings of the 5th Workshop on Representation Learning for NLP*, pages 120–130, Online, July 2020. Association for Computational Linguistics.

A Appendix

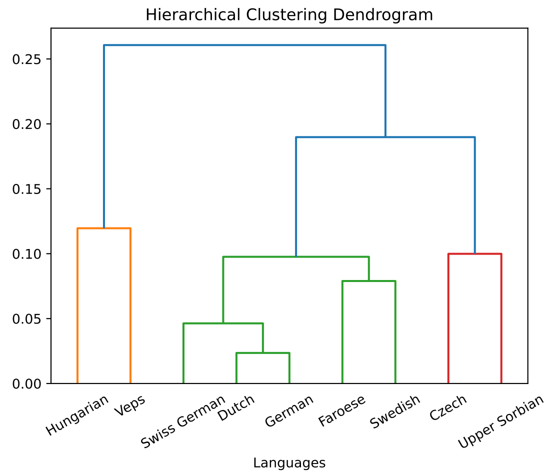


Fig. 1. Agglomerative Hierarchical Clustering dendrogram based on the cosine similarity of the `syntax_knn` vectors of the included languages.

B Appendix

Language	**Treebank**	**Number of tokens**	**Number of sentences**
Czech	CAC	494 142	24 709
Dutch	LassySmall	297 486	17 120
Faroese	OFT	10 002	1 208
German	HDT	3 399 390	189 928
Hungarian	Szeged	42 032	1 800
Swedish	Talbanken	96 859	6 026
Swiss German	ATB	652	98
Upper Sorbian	UFAL	11 196	646
Veps	VWT	1 303	103

Table 1. The specific UD treebanks featured in the project.